

Early Estimates of Cancer Incidence for 2015: Expanding to Include Estimates for White and Black Races

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BACKGROUND: The National Cancer Institute's cancer incidence estimates through 2015 from the Surveillance, Epidemiology, and End Results (SEER) registries' November 2017 submission are released in April 2018. **METHODS:** Early estimates (February 2017) of cancer incidence rates and trends from the SEER 18 registries for diagnoses in 2000 through 2015 were evaluated with a revised delay-adjustment model, which was used to adjust for the undercount of cases in the early release. For the first time, early estimates were produced for race (whites and blacks) along with estimates for new sites: the oral cavity and pharynx, leukemia, and myeloma. **RESULTS:** Model validation comparing delay-adjusted rates and trends through 2014 and using 2016 submissions showed good agreement. Differences in trends through 2015 in comparison with those through 2014 were evident. The rate of female breast cancer rose significantly from 2004 to 2015 by 0.3% per year (annual percent change [APC] = 0.3%); the prior trend through 2014 (the same magnitude) was not yet significant. The female colon and rectum cancer trend for whites became flat after previously declining. Lung and bronchus cancer for whites showed a significant decline (APC for males = -2.3%, 2012-2015; APC for females = -0.7%, 2011-2015). Thyroid cancer for black females changed from a continuous rise to a flat final segment (APC = 1.6%, not significant, 2011-2015). Both kidney and renal pelvis cancer (APC = 1.5%, 2011-2015) and childhood cancers (APC = 0.5%, 2000-2015) for white males showed a significant rise in the final segments from previously flat trends. Kidney and renal pelvis cancer for black males showed a change from a significant rise to a flat trend. **CONCLUSIONS:** The early release of SEER data continues to be useful as a preliminary estimate of the most current cancer incidence trends. *Cancer* 2018;124:2192-204. © 2018 American Cancer Society.

KEYWORDS: annual percent change, delay adjustment, early cancer rate estimates by race, early estimates of cancer incidence rates, population-based registry data.

INTRODUCTION

Using data reported to the Surveillance, Epidemiology, and End Results (SEER) program in February 2017, this report presents an early look at cancer incidence through the diagnosis year 2015. This article represents the preliminary look at trends in the latest SEER data before the usual release in April 2018 of data reported in November 2017. This is the fourth report in the series of early estimates of SEER data. The first 2 published reports^{1,2} set a foundation for evaluating early estimates of incidence rates and trends and introduced a modification of the usual November submission delay model to account for the larger undercount found in February submissions. The third report (with cases diagnosed from 2000 to 2014) for the diagnosis year 2014³ expanded the cancer sites that were evaluated and revised the delay model to make it more consistent with the reporting delay model used in the usual November submission. In addition to the usual reporting for all races combined, this article reports early estimates of cancer incidence separately for whites and blacks by sex for the first time. Slight revisions were made to the prior year's reporting delay model. To expand the reporting by cancer site, new cancer sites, including the oral cavity and pharynx, leukemia, and myeloma, have been added because these sites had prior increasing trends.

MATERIALS AND METHODS

Data Source and Methodology for Assessing Trends

Cancer incidence data from the SEER 18 registries were used in this analysis; they represent approximately 28% of the US population. The registries that comprise the SEER 18 include the San Francisco–Oakland, San Jose–Monterey, Los

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Angeles, greater California, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, Alaska Native, greater Georgia, rural Georgia, Kentucky, Louisiana, and New Jersey registries. These data were adjusted to reflect population changes from Hurricanes Katrina and Rita in 2005. Trends for cases diagnosed from 2000 through 2015 are reported on the basis of the February 2017 submission to the National Cancer Institute. Validation of the prior preliminary estimates and trends from 2000 through 2014 based on the February 2016 submission was conducted with the usual full November 2016 submission. The February 2017 submission represents the seventh February submission, and the reporting delay model was developed on the basis of February submissions from 2011 to 2017 plus the SEER full-data November submissions from 2002 to 2016.

The age-adjusted cancer incidence trends were assessed with the joinpoint regression model.³ The model allows up to 2 joinpoints for rates from 2000 to 2015, with 3 maximum segments. The resulting trends, reflecting the annual percent change (APC), were evaluated for each segment. The last segment often is of primary interest because it indicates whether the most current trend has changed. In addition, both 5- and 10-year year average annual percent changes (AAPCs)⁴ were evaluated for the trend magnitude and direction over fixed intervals.

Updates to the February Submission Reporting Delay Model

Reporting delay models have been used to adjust for undercounts of cases due to delays in the reporting of annual cancer incidence since the 2003 National Cancer Institute cancer statistics review.⁵ Before 2015, delay factors for SEER November submissions were based on prior data from November submissions from the SEER areas. In 2015 (with cases diagnosed through 2012), the SEER November 2015 submissions, reporting registry-specific delay factors, were estimated on the basis of December 2015 submissions of data to the North American Association of Central Cancer Registries (NAACCR); this takes place in December of each year. This allowed delay adjustments to be expanded to registries throughout the United States and Canada.

The idea behind modeling reporting delays is to adjust the current case count to account for anticipated future corrections (both additions and deletions) to the data. These adjusted counts and the associated delay model are valuable in more precisely determining current cancer trends and in monitoring the timeliness of data collection, which is an important aspect of quality control.^{4,6}

A reporting delay is estimated via the modeling of historical patterns of cases that have accumulated over many submissions for each diagnosis year (<https://surveillance.cancer.gov/delay/>). For a specific submission of the data, delay factors are assigned to each case, and they can vary by strata defined by cancer site, registry, age group, race, and year of diagnosis. These individual factors represent the amount that a specific case is weighted to account for the additional cases that will eventually be added to a specified stratum (eg, an individual delay factor of 1.04 means that 4% more cases should eventually be added to that stratum). Composite delay-adjusted rates are computed with SEER*Star⁷ via the weighing of each case by the delay factor for any desired calculation. Composite delay-adjustment factors (CDAFs) are computed by the division of the delay-adjusted rate by the corresponding observed non-delay-adjusted rate. For any particular submission, the delay factors are largest for the most recent diagnosis years because they have the largest undercount of cases. This differential undercount of cases in the most recent diagnosis years can cause a bias when one is estimating the most recent trends with observed data, even when the size of the delay factors is modest. Although end-of-year submissions (November for SEER and December for NAACCR) are available for every registry, delay factors based on these submissions are estimated by registry throughout the United States and Canada. Because February submissions are available only from SEER registries, delay factors for these submissions could be estimated only for the SEER 18 areas. To make it more consistent with the model used for the NAACCR December 2015 submission, several new features were introduced into the delay model used with the February 2016 SEER submission. Modifications included the fact that delay factors were estimated by registry and cancer site and could vary by race, age group (<50, 50-64, and ≥65 years), and sex. The determination of how the delay factors varied across these strata depended on a minimum sample size of 100 for each site, race, age group, and sex marginal total (with pooling if cells were too small). If the average number of cases per year for a cancer site across all racial, age, and sex groups was less than 100, then the delay factors were estimated on the basis of all the SEER registries combined.

Additional modifications to the delay model were introduced for use with the NAACCR December 2016 submission, and corresponding changes were made for use with the February 2017 delay model. Changes included the following: 1) sex was dropped in the delay model because of its small impact on the result, 2) the threshold for collapsing over covariate groups was

changed from 100 to 25 so that fewer racial and age groups would be merged because of small sample sizes, and 3) the determination of how the delay factors varied across these strata was based on a slightly different procedure to combine strata that did not have sufficient counts.

Model Validation

In previous reports,^{1,2} we presented validation tables with delay-adjusted rates and delay-adjusted trends from the SEER 18 registries; these started in 2000 from a February submission and a subsequent November submission in the same calendar year. The purpose of the validation was to compare November-to-February rate ratios for the observed and delay-adjusted incidence rates to determine whether the delay adjustment produced rates from the 2 submissions that were comparable. In addition, joinpoint trends from the February submission and the standard November submission were compared.

The ratios of the observed rates are usually substantially greater than 1, and this reflects substantial underreporting in the February submission in comparison with the subsequent November submission for the most recent diagnosis year. However, if the delay adjustment is working properly, the ratio of the November submissions to the prior February submissions of the delay-adjusted rates should be close to 1 because both are adjusted to the eventual counts that should be observed after many submissions. In addition, APC and 5- and 10-year AAPC trends, estimated from joinpoint models fit to the data from the February submission, were validated against trends fit to the subsequent November submissions. In the past, these validations demonstrated that the rates and trends estimated from the February submissions were reasonably close to those estimated from the subsequent November submissions.

Comparison of 2014 rates with the February 2016 and November 2016 submissions

The validation tables are included for all races and, for the first time, for whites and blacks (Supporting Tables 1, 2, and 3, respectively). The all-race observed ratios of 2014 incidence rates for the November 2016 submission versus the February 2016 submission ranged from 1.036 to 1.163 (median, 1.076), and this indicated a median undercount of 7.6% for the February submission relative to the subsequent November submission. The all-race delay-adjusted model corrected for the February undercount, with the November-to-February delay-adjusted rate ratios ranging from 0.984 to 1.071 (median, 1.014); this indicated a median undercount of 1.4%. The cancer

site with the largest delay-adjusted rate ratios for all races was melanoma of skin for males (1.071) and females (1.053), but these ratios were much larger before the delay adjustment (1.155 and 1.129, respectively). Other sites with higher delay-adjusted ratios included the prostate (1.040), all cancer sites for females aged 0 to 19 years (1.032), and the liver and intrahepatic bile duct (IBD) for males and females (both ratios were 1.028). The ratios for white race for the observed 2014 incidence rates for the November 2016 to February 2016 submissions ranged from 1.039 to 1.163 (median, 1.087), and they ranged from 0.985 to 1.062 (median, 1.014) for delay-adjusted rates. Both ratios were very similar to the all-race ratios. The ratio ranges for black race were 1.001 to 1.142 (median, 1.074) for the observed rates (Supporting Table 3) and 0.977 to 1.067 (median, 1.010) for the delay-adjusted rates. The ranges of ratios for delay-adjusted rates for black race were comparable to those for whites and all races.

Comparison of 2014 trends with the February 2016 and November 2016 submissions

The all-race APC trends and 5- and 10-year AAPC trends were nearly identical for most sites (Supporting Table 1). Exceptions included the APC trends for male colon and rectum cancer (CRC) for all races, which had no joinpoints from the February 2016 submission but had 2 joinpoints in 2008 and 2012 in the November 2016 submission, with the November submission showing a flattening of the prior decline in incidence.

APC trends and 5- and 10-year AAPC trends for white and black race revealed slightly more inconsistencies between the 2 submissions than were observed for all races (Supporting Tables 2 and 3). Among the 28 series analyzed, differences in the number of joinpoints occurred for 1 series for all races, for 4 series for whites, and for 4 series for blacks. For all races, male CRC went from no joinpoint to 2 joinpoints. For whites, male and female lung and bronchus cancer each went from 1 joinpoint to 2 joinpoints, with the final segment showing a flattening based on the November submission. Cervical cancer trends went from 2 joinpoints to 1 joinpoint but still showed a statistically significant decline in both submissions. Female leukemia went from 1 joinpoint to no joinpoints but showed a significant decline in the final segment in both submissions. Among blacks, female CRC went from 1 joinpoint to no joinpoints, but the final segment still showed a significant decline. Female liver/IBD cancer changed from 1 joinpoint to no joinpoints but still showed a significant increase in the final segments.

Joinpoints in the trends for male and female kidney and renal pelvis cancer changed but went from 1 joinpoint to no joinpoints for males and from 1 joinpoint to 2 joinpoints for females.

RESULTS

Composite Delay-Adjustment Factors (CDAFs)

To estimate the delay factors of the aggregates of these groups, a CDAF is estimated by the division of the delay-adjusted rate by the observed rate. To assist with the interpretation of the CDAFs, it should be noted that these vary with submission date (February vs November), cancer-specific site, and race (white, black, and all races). Delay factors vary by cancer site because of differences in how cancers are diagnosed and treated and the corresponding footprint that they leave on the medical care system. For example, chronic lymphocytic leukemia is usually diagnosed in a doctor's office through blood tests and is often monitored rather than treated immediately. This minimal footprint in comparison with other cancers causes a delay in identifying these cases, and this contributes to the large CDAFs for leukemias. Race-specific delay factors are larger than all-race combined factors because cases are often identified and entered into the registry but only later receive a race specification. February submissions for the most recent diagnosis years have significantly larger delay factors than the subsequent November submissions because they allow the registry an additional 9 months to identify cases. Although standard errors for CDAFs are not regularly computed (they instead are computed for delay-adjusted rates to be used in trend analysis), prior methodological work has indicated that almost all CDAFs are statistically different than 1 for individual registries and, unless estimated for a very rare cancer and/or a very small subpopulation, are statistically different than 1 when pooled across groups of registries.

The observed and delay-adjusted incidence rates for diagnosis year 2015 from the SEER February 2017 data submission are compared with those for diagnosis year 2014 in Tables 1, 2, and 3 for all races, white race, and black race, respectively. The 2014 delay-adjusted rates are the same as those reported by the SEER program in the Cancer Statistics Review, which was published in April 2017.⁸ The CDAFs based on the 2014 age-adjusted rates from the November 2016 submission and the 2015 age-adjusted rates from the February 2017 submission are shown. The CDAFs for the February 2017 submission ranged from 1.04 for corpus and uterus cancer not otherwise specified (NOS) to 1.28 for male leukemia (median,

1.10) for all races, whereas those for whites ranged from 1.05 for corpus and uterus cancer NOS and female breast cancer to 1.31 for male leukemia (median, 1.11), and those for blacks ranged from 1.06 for corpus and uterus cancer NOS to 1.32 for male leukemia (median, 1.12). Overall, the November 2016 CDAFs were smaller than the February 2017 CDAFs, which also had more variability than the ones from the November submission. The CDAFs for the November 2016 submission ranged from 1.01 for corpus and uterus cancer NOS to 1.13 for male and female leukemia (median, 1.03) for all races, whereas those for whites ranged from 1.01 for corpus and uterus cancer NOS to 1.15 for male and female leukemia (median, 1.04), and those for blacks ranged from 1.03 for corpus and uterus cancer NOS to 1.16 for male leukemia (median, 1.05). There were large delay-adjustment factors for lung and bronchus cancer for both men and women (Figs. 1 and 2). The 2015 non-delay-adjusted, age-adjusted rate for 2015 was very low. However, after the adjustment, the trends through 2015 were consistent with the trends that were observed through 2014. The CDAFs for the February 2017 submission ranged from 1.04 for corpus and uterus cancer to 1.28 for male leukemia for all races (median, 1.10). CDAFs among whites ranged from 1.05 for female breast cancer and corpus and uterus cancer to 1.31 for male leukemia (median, 1.11). CDAFs among blacks ranged from 1.06 for corpus and uterus cancer to 1.32 for male leukemia (median, 1.12).

Joinpoint Trends Through 2015 for All Races

Table 1 compares APC trends and 5- and 10-year AAPC trends for all races. Figures 1 and 2 show delay-adjusted and non-delay-adjusted trends from 2000 to 2015 from the February 2015 submission for selected cancer sites for males and females, respectively. Male CRC for all races showed some preliminary signs of decreasing more rapidly from 2012 to 2015 (APC = -0.9%, not significant) in comparison with the final segment from the November submission (APC = -0.2%, not significant, 2012-2014). For female breast cancer, a previous nonsignificant trend for the last segment (APC = 0.3%, 2004-2014) became significant after the addition of the 2015 delay-adjusted data point (APC = 0.3%, 2004-2015). The all-race prostate cancer trend continued to decline significantly (APC = -7.2%, 2009-2015; middle left graph in Fig. 1). The trend for female kidney and renal pelvis cancer (top right graph in Fig. 2) showed a nonsignificant rise in the final segment from 2007 to 2015 (APC = 0.05%, not significant) on the basis of the February 2017 submission, whereas the November 2016 submission had a final

TABLE 1. Comparison of November 2016 (Through Diagnosis Year 2014) and February 2017 (Through Diagnosis Year 2015) SEER 18 Submissions: All Races

Cancer Sites	November 2016 Submission				February 2017 Submission			
	2014 Age-Adjusted Rates		CDAF	APC Trends	2015 Age-Adjusted Rates		CDAF	APC Trends
	Observed	Delay-Adjusted			Observed	Delay-Adjusted		
All sites, male	460.2	475.0	1.03	-0.7 ^a (2000-2009) -2.6 ^a (2009-2014)	421.8	463.0	1.10	-0.7 ^a (2000-2009) -2.6 ^a (2009-2015)
All sites, female	406.1	418.9	1.03	-0.1 (2000-2014)	385.4	421.5	1.09	-0.0 (2000-2015)
Colon and rectum, male	44.4	45.3	1.02	-2.5 ^a (2000-2008) -3.9 ^a (2008-2012) -0.2 (2012-2014)	40.9	43.7	1.07	-2.5 ^a (2000-2008) -3.8 ^a (2008-2012) -0.9 (2012-2015)
Colon and rectum, female	33.9	34.6	1.02	-2.4 ^a (2000-2014)	31.4	33.6	1.07	-2.4 ^a (2000-2015)
Lung and bronchus, male	59.3	61.1	1.03	-1.9 ^a (2000-2008) -3.2 ^a (2008-2014)	53.1	58.9	1.11	-1.9 ^a (2000-2008) -3.1 ^a (2008-2015)
Lung and bronchus, female	46.3	47.7	1.03	-0.0 (2000-2007) -1.7 ^a (2007-2014)	42.1	46.7	1.11	0.2 (2000-2006) -1.5 ^a (2006-2015)
Melanoma, male	30.3	31.2	1.03	1.9 ^a (2000-2014)	26.0	29.0	1.12	1.7 ^a (2000-2015)
Melanoma, female	18.2	18.7	1.03	1.3 ^a (2000-2014)	16.1	18.0	1.11	1.3 ^a (2000-2015)
Female breast	125.1	127.2	1.02	-2.3 ^a (2000-2004) 0.3 (2004-2014)	122.2	127.9	1.05	-2.3 ^a (2000-2004) 0.3 ^a (2004-2015)
Prostate	97.3	101.4	1.04	-1.9 ^a (2000-2010) -9.2 ^a (2010-2014)	90.7	102.2	1.13	-1.6 ^a (2000-2009) -7.2 ^a (2009-2015)
Cervix	7.3	7.4	1.02	-3.8 ^a (2000-2003) -1.3 ^a (2003-2014)	7.0	7.5	1.06	-4.0 ^a (2000-2003) -1.2 ^a (2003-2015)
Thyroid, male	7.4	7.5	1.02	6.2 ^a (2000-2009) 2.6 ^a (2009-2014)	7.4	7.8	1.05	6.3 ^a (2000-2009) 2.5 ^a (2009-2015)
Thyroid, female	21.9	22.3	1.02	6.4 ^a (2000-2006) 7.9 ^a (2006-2009)	21.2	22.3	1.05	6.4 ^a (2000-2006) 8.1 ^a (2006-2009)
Liver/IBD, male	13.6	14.3	1.06	2.0 ^a (2009-2014) 4.1 ^a (2000-2009)	12.1	14.5	1.20	1.7 ^a (2009-2015) 4.1 ^a (2000-2009)
Liver/IBD, female	4.9	5.2	1.06	2.1 ^a (2009-2014)	4.2	5.0	1.20	2.0 ^a (2009-2015)
Pancreas, male	14.3	14.8	1.03	3.1 ^a (2000-2014)	13.0	14.8	1.14	3.1 ^a (2000-2015)
Pancreas, female	11.1	11.5	1.03	0.9 ^a (2000-2014)	10.3	11.7	1.14	0.9 ^a (2000-2015)
Kidney and renal pelvis, male	21.3	22.0	1.04	3.1 ^a (2000-2008) 0.2 (2008-2014)	20.6	22.6	1.10	3.0 ^a (2000-2008) 0.4 (2008-2015)
Kidney and renal pelvis, female	10.5	10.9	1.04	3.5 ^a (2000-2008) -0.3 (2008-2014)	10.6	11.6	1.10	3.7 ^a (2000-2007) 0.5 (2007-2015)
Corpus and uterus NOS	26.6	26.9	1.01	-1.4 (2000-2003) 1.4 ^a (2003-2014)	26.0	27.0	1.04	-1.4 (2000-2003) 1.4 ^a (2003-2015)
All sites (age, 0-19 y), male	18.9	19.4	1.03	0.6 ^a (2000-2014)	18.5	19.8	1.07	0.7 ^a (2000-2015)
All sites (age, 0-19 y), female	17.7	18.2	1.03	1.0 ^a (2000-2014)	17.6	18.9	1.07	1.1 ^a (2000-2015)
Oral cavity and pharynx, male	16.9	17.4	1.03	-0.5 (2000-2005) 1.1 ^a (2005-2014)	15.8	17.2	1.08	-0.4 (2000-2005) 1.0 ^a (2005-2015)
Oral cavity and pharynx, female	6.2	6.3	1.03	0.0 (2000-2014)	5.9	6.4	1.08	0.1 (2000-2015)

TABLE 1. Continued

Cancer Sites	November 2016 Submission					February 2017 Submission				
	2014 Age-Adjusted Rates			2015 Age-Adjusted Rates		2015 Age-Adjusted Rates			AAPC Trends: 5 y/10 y	
	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y
Leukemia, male	17.0	19.3	1.13	-0.4 (2000-2006) 1.6 ^a (2006-2014)	1.6 ^a /1.4 ^a	15.2	19.5	1.28	-0.4 (2000-2006) 1.6 ^a (2006-2015)	1.6 ^a /1.6 ^a
Leukemia, female	10.4	11.8	1.13	-0.3 (2000-2005) 1.6 ^a (2005-2014)	1.6 ^a /1.6 ^a	9.4	11.9	1.26	-0.3 (2000-2005) 1.6 ^a (2005-2015)	1.6 ^a /1.6 ^a
Myeloma, male	8.0	8.9	1.11	1.7 ^a (2000-2014)	1.7 ^a /1.7 ^a	7.4	9.0	1.23	1.7 ^a (2000-2015)	1.7 ^a /1.7 ^a
Myeloma, female	5.0	5.5	1.11	1.1 ^a (2000-2014)	1.1 ^a /1.1 ^a	4.8	5.9	1.22	1.3 ^a (2000-2015)	1.3 ^a /1.3 ^a

Abbreviations: AAPC, average annual percent change; APC, annual percent change; CDAF, composite delay-adjustment factor; IBD, intrahepatic bile duct; NOS, not otherwise specified; SEER, Surveillance, Epidemiology, and End Results.

^aAPC and AAPC trends are significant ($P < .05$).

segment with a modest nonsignificant increase (APC = 0.3%, not significant, 2008-2014). Other cancer sites showed no major new trends when the preliminary 2015 data point was added.

Joinpoint Trends Through 2015 for Whites and Blacks

Early estimates of cancer incidence for white and black race appear in Tables 2 and 3, respectively. Trends for white males generally stayed in the same direction for cases diagnosed through 2014 from the November 2016 submission and for cases diagnosed through 2015 from the February 2017 submission. One slight change was a downward trend for CRC among white males (APC = -0.4%, not significant, 2012-2015) from the February 2017 submission from a previously increasing trend (APC = 0.4%, not significant, 2012-2014) from the November 2016 submission. For white males, liver and IBD cancer had a slowing but still increasing trend (APC = 2.5%, significant, 2009-2015) in comparison with the trend through 2014 (APC = 3.5%, significant, 2005-2014; Fig. 1). The direction of the trend for white male kidney and renal pelvis cancer changed from being flat (APC = 0.2%, nonsignificant, 2008-2014) to a significant increase (APC = 1.5%, significant, 2011-2015), and there was a change from 2 joinpoint segments to 3 joinpoint segments. For white females (Fig. 2), the CRC trend was notable because of the change from a significant declining continuous trend (APC = -2.4%, significant, 2000-2014) to a trend with 2 joinpoints and 3 trend segments, with the last segment reflecting a slower albeit nonsignificant decrease (APC = -1.1%, 2011-2015; Fig. 2).

The trend for kidney and renal pelvis cancer among black males flattened (APC = -0.1%, not significant, 2009-2015) in contrast to the prior statistically significant decline of 2.2% from 2000 to 2014. The more recent trend for CRC for black females from 2007 to 2015 demonstrated a slightly steeper decline (APC = -3.4%, significant, 2007-2015) from the previous APC estimate of -2.5% (significant) for the period 2000-2014 (Table 3 and Fig. 2). For thyroid cancer among black females, the prior trend, which was increasing (APC = 5.9%, significant, 2000-2014), seems to have flattened (APC = 1.6%, not significant, 2011-2015; Table 3 and Fig. 2). For liver/IBD cancer among black females, the dramatic increase that was previously observed (APC = 3.2%, significant, 2000-2014) changed to a slight, nonsignificant increase in the latest data (APC = 0.9%, not significant, 2008-2015; Table 3 and Fig. 2).

TABLE 2. Comparison of November 2016 (Through Diagnosis Year 2014) and February 2017 (Through Diagnosis Year 2015) SEER 18 Submissions: White Race

Cancer Site	November 2016 Submission					February 2017 Submission				
	2014 Age-Adjusted Rates			2015 Age-Adjusted Rates		2015 Age-Adjusted Rates			2015 Age-Adjusted Rates	
	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y
All sites, male	466.5	483.9	1.04	-0.6 ^a (2000-2009) -2.6 ^a (2009-2014)	-2.6 ^a /-1.7 ^a	425.1	472.2	1.11	-0.6 ^a (2000-2009) -2.5 ^a (2009-2015)	-2.5 ^a /-1.9 ^a
All sites, female	419.6	434.8	1.04	-0.0 (2000-2014)	-0.0/-0.0	397.7	440.0	1.11	0.0 (2000-2015)	0.0/0.0
Colon and rectum, male	43.9	44.8	1.02	-2.6 ^a (2000-2008) -4.1 ^a (2008-2012) 0.4 (2012-2014)	-1.9 ^a /-2.7 ^a	40.3	43.3	1.07	-2.6 ^a (2000-2008) -4.0 ^a (2008-2012) -0.4 (2012-2015)	-1.4/-2.5 ^a
Colon and rectum, female	33.6	34.3	1.02	-2.4 ^a (2000-2014)	-2.4 ^a /-2.4 ^a	31.2	33.5	1.07	-2.2 ^a (2000-2008) -3.6 ^a (2008-2011) -1.1 (2011-2015)	-1.1/-2.2 ^a
Lung and bronchus, male	59.7	61.5	1.03	-1.9 ^a (2000-2009) -4.0 ^a (2009-2012) -1.9 (2012-2014)	-2.9 ^a /-2.6 ^a	53.2	59.2	1.11	-1.9 ^a (2000-2009) -3.9 ^a (2009-2012) -2.3 ^a (2012-2015)	-2.7 ^a /-2.7 ^a
Lung and bronchus, female	48.6	50.0	1.03	0.1 (2000-2007) -2.1 ^a (2007-2012) -0.1 (2012-2014)	-1.1/-1.2 ^a	44.3	49.3	1.11	-0.0 (2000-2008) -2.8 ^a (2008-2011) -0.7 ^a (2011-2015)	-0.7 ^a /-1.3 ^a
Melanoma, male	35.4	37.2	1.05	2.1 ^a (2000-2014)	2.1 ^a /2.1 ^a	30.5	35.1	1.15	1.9 ^a (2000-2015)	1.9 ^a /1.9 ^a
Melanoma, female	22.0	23.0	1.05	1.6 ^a (2000-2014)	1.6 ^a /1.6 ^a	19.4	22.3	1.15	1.5 ^a (2000-2015)	1.5 ^a /1.5 ^a
Female breast	127.9	129.9	1.02	-2.5 ^a (2000-2004) 0.2 (2004-2014)	0.2/0.2	124.5	131.0	1.05	-2.5 ^a (2000-2004) 0.2 (2004-2015)	0.2/0.2
Prostate	91.3	97.3	1.07	-1.9 ^a (2000-2010) -9.5 ^a (2010-2014) -3.9 ^a (2000-2003) -1.1 ^a (2003-2014)	-9.5 ^a /-5.4 ^a	83.0	96.9	1.17	-1.6 ^a (2000-2009) -7.5 ^a (2009-2015) -1.4 ^a (2000-2015)	-7.5 ^a /-5.6 ^a
Cervix	7.3	7.5	1.02	6.4 ^a (2000-2009) 2.3 ^a (2009-2014)	-1.1 ^a /-1.1 ^a	7.1	7.6	1.07	6.4 ^a (2000-2009) 2.3 ^a (2009-2015)	-1.4 ^a /-1.4 ^a
Thyroid, male	7.7	7.9	1.02	2.3 ^a (2000-2009) 7.1 ^a (2000-2009) 2.1 ^a (2009-2014)	2.3 ^a /4.1 ^a	7.9	8.3	1.06	7.2 ^a (2000-2009) 1.8 ^a (2009-2015)	2.3 ^a /3.7 ^a
Thyroid, female	23.2	23.7	1.02	-1.4 (2000-2002) 7.5 ^a (2002-2005) 3.5 ^a (2005-2014) -0.3 (2000-2003) 4.3 ^a (2003-2014)	3.5 ^a /3.5 ^a	22.2	23.5	1.06	2.3 ^a (2000-2009) 7.2 ^a (2000-2009) 1.8 ^a (2009-2015)	1.8 ^a /3.5 ^a
Liver/IBD, male	12.3	13.0	1.06	10.0 (2000-2014)	10.0/1.0 ^a	10.7	12.9	1.21	10.0 (2000-2015) 3.2 ^a (2000-2008) -0.7 (2008-2011)	2.5 ^a /3.4 ^a
Liver/IBD, female	4.5	4.7	1.06	10.0 (2000-2014)	10.0/1.0 ^a	3.8	4.6	1.21	10.0 (2000-2015) 3.2 ^a (2000-2008) -0.7 (2008-2011)	4.2 ^a /4.2 ^a
Pancreas, male	14.4	14.8	1.03	10.0 (2000-2014)	10.0/1.0 ^a	13.2	15.2	1.15	10.0 (2000-2015) 3.2 ^a (2000-2008) -0.7 (2008-2011)	1.0 ^a /1.0 ^a
Pancreas, female	11.1	11.4	1.03	10.0 (2000-2014)	10.0/1.0 ^a	10.2	11.7	1.14	10.0 (2000-2015) 3.2 ^a (2000-2008) -0.7 (2008-2011)	1.1 ^a /1.1 ^a
Kidney and renal pelvis, male	21.9	22.7	1.04	3.0 ^a (2000-2008) 0.2 (2008-2014)	0.2/1.2 ^a	21.2	23.3	1.10	3.2 ^a (2000-2008) -0.7 (2008-2011) 1.5 ^a (2011-2015)	1.5 ^a /1.1 ^a
Kidney and renal pelvis, female	10.9	11.3	1.04	3.9 ^a (2000-2007) 0.1 (2007-2014) -1.7 (2000-2003) 1.3 ^a (2003-2014)	0.1/0.9 ^a	11.1	12.2	1.10	3.8 ^a (2000-2007) 0.6 (2007-2015) -1.7 (2000-2003) 1.3 ^a (2003-2015)	0.6/0.9 ^a
Corpus and uterus NOS	27.4	27.7	1.01	3.9 ^a (2000-2007) 0.1 (2007-2014) -1.7 (2000-2003) 1.3 ^a (2003-2014)	1.3 ^a /1.3 ^a	26.7	27.9	1.05	3.8 ^a (2000-2007) 0.6 (2007-2015) -1.7 (2000-2003) 1.3 ^a (2003-2015)	1.3 ^a /1.3 ^a

TABLE 2. Continued

Cancer Site	November 2016 Submission					February 2017 Submission				
	2014 Age-Adjusted Rates			2015 Age-Adjusted Rates		2015 Age-Adjusted Rates			2015 Age-Adjusted Rates	
	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y	Observed	Delay-Adjusted	CDAF	APC Trends	AAPC Trends: 5 y/10 y
All sites (age, 0-19 y), male	19.9	20.5	1.03	0.4 (2000-2014)	0.4/0.4	19.6	21.2	1.08	0.5 ^a (2000-2015)	0.5 ^a /0.5 ^a
All sites (age, 0-19 y), female	18.5	19.0	1.03	0.9 ^a (2000-2014)	0.9 ^a /0.9 ^a	18.9	20.5	1.08	1.1 ^a (2000-2015)	1.1 ^a /1.1 ^a
Oral cavity and pharynx, male	17.7	18.3	1.04	-0.7 (2000-2003)	1.3 ^a /1.3 ^a	16.7	18.3	1.10	-0.6 (2000-2003)	1.3 ^a /1.3 ^a
Oral cavity and pharynx, female	6.4	6.6	1.04	1.3 ^a (2003-2014)	0.4 ^a /0.4 ^a	5.9	6.5	1.10	1.3 ^a (2003-2015)	0.3 ^a /0.3 ^a
Leukemia, male	17.9	20.6	1.15	0.4 ^a (2000-2014)	1.7 ^a /1.5 ^a	16.0	20.9	1.31	0.3 ^a (2000-2015)	1.6 ^a /1.6 ^a
Leukemia, female	10.9	12.6	1.15	-0.2 (2000-2006)	1.7 ^a (2006-2014)	9.8	12.7	1.29	-0.2 (2000-2006)	1.2 ^a /1.2 ^a
Myeloma, male	7.6	8.5	1.12	1.2 ^a (2000-2014)	1.2 ^a /1.2 ^a	7.0	8.7	1.24	1.2 ^a (2000-2015)	1.8 ^a /1.8 ^a
Myeloma, female	4.5	5.0	1.12	1.7 ^a (2000-2014)	1.7 ^a /1.7 ^a	4.4	5.4	1.24	1.8 ^a (2000-2015)	1.3 ^a /1.3 ^a

Abbreviations: AAPC, average annual percent change; APC, annual percent change; CDAF, composite delay-adjustment factor; IBD, intrahepatic bile duct; NOS, not otherwise specified; SEER, Surveillance, Epidemiology, and End Results.

^aAPC and AAPC trends are significant ($P < .05$).

DISCUSSION

This is the fourth report on the use of improved delay-adjustment models to provide early estimates for a variety of cancer sites. We have added 3 new cancer sites (oral cavity and pharynx, leukemia, and myeloma) this year, and we have added estimates for white and black race for the first time. This report includes early estimates of cancer incidence for the 2015 diagnosis year for 16 cancer sites based on the SEER February 2017 data submission.

The rate ratios of the delay-adjusted rates from the November 2016 submission to the February 2016 submission were relatively close to 1 for all races, white race, and black race, and this gives confidence that the delay-adjustment factors for the February 2016 submission are performing well. As for the validation of trends, out of the cancer combinations included in this study, there were only 3 combinations among whites and 4 among blacks for which the number of joinpoints changed when we compared trends from 2000 to 2014 based on the February 2016 and November 2016 submissions. The delay-adjustment model for the February submissions continues to provide relatively consistent results for incidence rate estimates and APC trends for the February 2016 submission in comparison with the November 2016 submission. Although white and black race estimates were added this year, in future years, adding Asian-Pacific Islander to the race specification, Hispanic ethnicity, and race/ethnicity combinations will be considered. The NAACCR December submission delay model has already been extended to these groups, but these models would need to be extended for the February submission model.

Several sites with potentially important changes in trends according to the preliminary data through the 2015 diagnosis year include the colon and rectum, lung and bronchus, cervix, thyroid, and kidney and renal pelvis. For whites, the lung and bronchus decline became significant in the most recent time period through 2015, whereas the downward trend among blacks continued. Other important changes to track include the continuing decline of prostate cancer. These are noted in Figures 1 and 2. Monitoring new changes in trends can provide insight into areas where cancer control efforts may be working and indicate cancer sites that need continuing attention. New findings from the estimated trends through 2015 indicate a continuing decline in CRC for males of all races, with rates beginning to plateau in 2012; this is similar to white male rates. Black male rates are continuing to decline with no apparent change in trend (APC = -2.48%, significant, 2000-2015). Incidence trends of CRC for white females and black females continue to

TABLE 3. Comparison of November 2016 (Through Diagnosis Year 2014) and February 2017 (Through Diagnosis Year 2015) SEER 18 Submissions: Black Race

Cancer Site	November 2016 Submission				February 2017 Submission			
	2014 Age-Adjusted Rates			AAPC Trends: 5 y/10 y	2015 Age-Adjusted Rates			AAPC Trends: 5 y/10 y
	Observed	Delay-Adjusted	CDAF		APC Trends	Observed	Delay-Adjusted	
All sites, male	492.6	522.4	1.06	−3.9 ^y /−2.3 ^a	443.5	501.3	1.13	−1.8 ^a (2000-2006) 0.2 (2006-2009) −3.9 ^a (2009-2015)
	390.3	413.8	1.06	0.1/0.1	367.5	414.2	1.13	0.1 (2000-2015)
Colon and rectum, male	49.6	51.8	1.04	−3.8 ^y /−3.1 ^a	49.7	54.5	1.10	−2.5 ^a (2000-2015)
Colon and rectum, female	40.0	41.8	1.04	−2.5 ^y /−2.5 ^a	36.3	39.8	1.10	−1.6 ^a (2000-2007) −3.4 ^a (2007-2015)
Lung and bronchus, male	76.1	79.1	1.04	−2.8 ^y /−2.8 ^a	66.0	73.6	1.12	−2.9 ^a (2000-2015)
Lung and bronchus, female	48.6	50.5	1.04	−1.4 ^y /−1.4 ^a	43.8	48.8	1.11	1.4 (2000-2005) −1.5 ^a (2005-2015)
Melanoma, male	1.2	1.3	1.09	−0.4/−0.4	1.2	1.4	1.13	−0.1 (2000-2015)
Melanoma, female	1.0	1.1	1.08	0.9/0.9	1.0	1.2	1.14	0.9 (2000-2015)
Female breast	125.4	130.5	1.04	0.6 ^y /0.6 ^a	122.1	130.5	1.07	0.6 ^a (2000-2015)
Prostate	149.6	163.7	1.09	−7.9 ^y /−4.7 ^a	135.2	161.8	1.20	−1.9 ^a (2000-2010) −7.3 ^y /−4.9 ^a
Cervix	8.1	8.4	1.03	−2.6 ^y /−2.6 ^a	7.2	7.9	1.09	−2.7 ^y /−2.7 ^a
Thyroid, male	4.4	4.5	1.04	4.5 ^a /4.5 ^a	3.6	3.9	1.08	4.1 ^a /4.1 ^a
Thyroid, female	14.6	15.1	1.03	5.9 ^a /5.9 ^a	13.2	14.1	1.07	6.7 ^a (2000-2011) 1.6 (2011-2015)
Liver/IBD, male	17.5	18.5	1.06	3.4 ^a /3.4 ^a	16.4	19.7	1.20	3.4 ^a (2000-2015)
Liver/IBD, female	5.1	5.3	1.06	3.2 ^a /3.2 ^a	4.4	5.3	1.20	5.1 ^a (2000-2008) 0.9 (2008-2015)
Pancreas, male	17.8	18.5	1.04	0.5/0.5	14.7	16.9	1.15	0.4 (2000-2015)
Pancreas, female	14.0	14.6	1.04	−0.7/−0.7	12.7	14.6	1.15	2.5 (2000-2005) −0.6 (2005-2015)
Kidney and renal pelvis, male	25.7	27.0	1.05	2.2 ^a /2.2 ^a	23.7	26.4	1.11	3.5 ^a (2000-2009) −0.1 (2009-2015)
Kidney and renal pelvis, female	12.1	12.7	1.05	−0.8/2.1	11.6	12.9	1.11	2.1 (2000-2005) 8.0 (2005-2008) −0.7 (2008-2015)
Corpus and uterus NOS	25.4	26.1	1.03	2.4 ^a /2.4 ^a	24.9	26.4	1.06	2.3 ^a (2000-2015)
All sites (age, 0-19 y), male	13.3	14.1	1.06	1.5 ^y /1.5 ^a	12.2	13.5	1.10	1.3 ^y /1.3 ^a
All sites (age, 0-19 y), female	14.0	14.8	1.06	1.8 ^a /1.8 ^a	10.3	11.3	1.10	1.3 ^a (2000-2015)
Oral cavity and pharynx, male	13.9	14.5	1.04	−2.2 ^a /−2.2 ^a	12.0	13.2	1.10	−2.3 ^a (2000-2015)
Oral cavity and pharynx, female	4.9	5.1	1.04	−1.3 ^y /−1.3 ^a	4.7	5.2	1.10	−1.2 ^y /−1.2 ^a
Leukemia, male	13.4	15.5	1.16	0.7/0.7	11.5	15.1	1.32	0.7 (2000-2015)
Leukemia, female	8.7	10.0	1.15	2.7 ^y /1.8 ^a	7.7	10.1	1.30	−1.2 (2000-2007) 2.6 ^a (2007-2015)
Myeloma, male	14.5	16.5	1.14	1.4 ^a /1.4 ^a	12.9	16.4	1.26	1.2 ^a (2000-2015)
Myeloma, female	9.8	11.1	1.14	1.2 ^a /1.2 ^a	10.0	12.6	1.26	1.3 ^a (2000-2015)

Abbreviations: AAPC, average annual percent change; APC, annual percent change; CDAF, composite delay-adjustment factor; IBD, intrahepatic bile duct; NOS, not otherwise specified; SEER, Surveillance, Epidemiology, and End Results.

^a APC and AAPC trends are significant ($P < .05$).

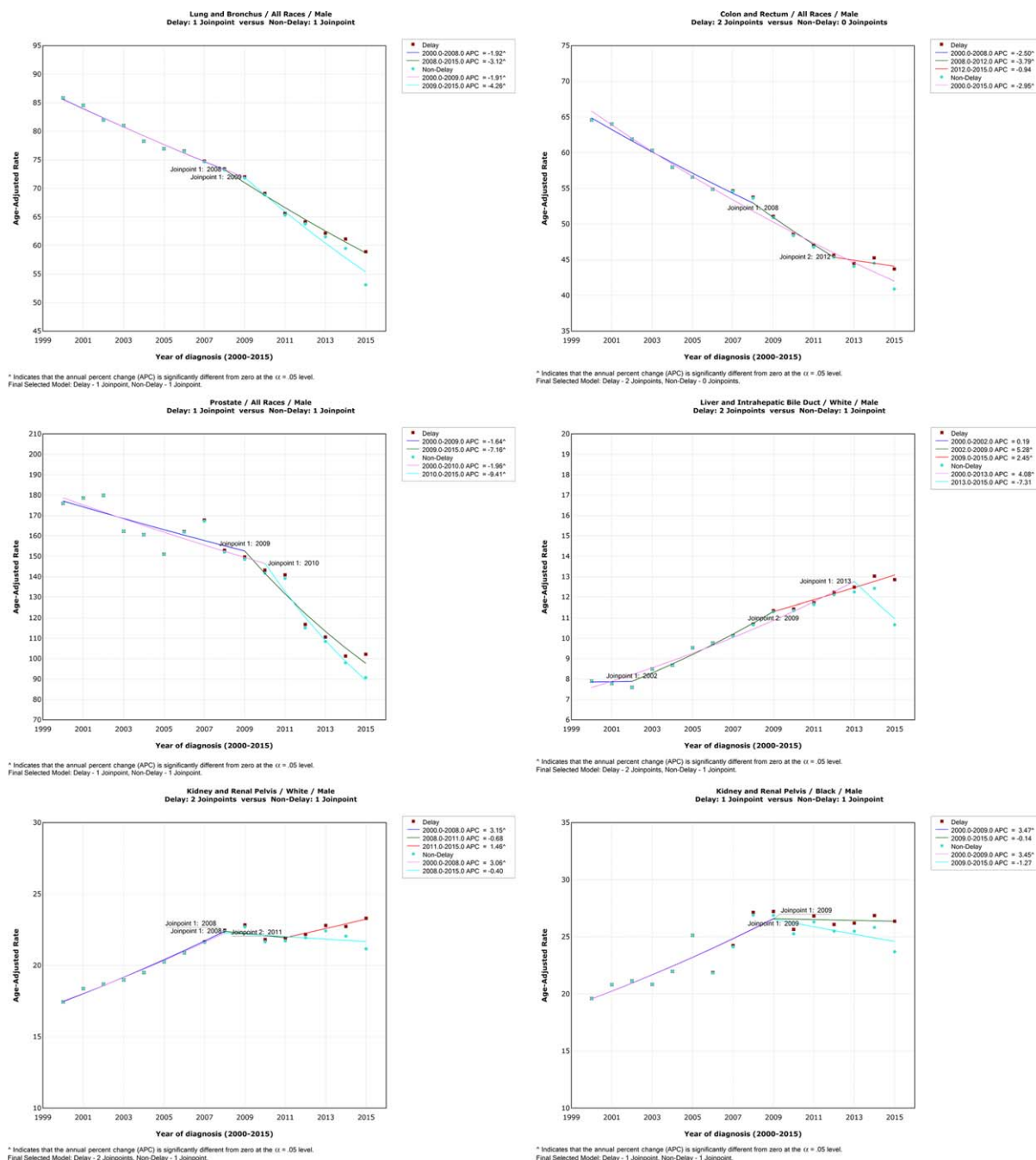


Figure 1. SEER 18 registries' observed and delay-adjusted incidence rates for selected sites for males for the diagnosis years 2000-2015 from the February 2017 SEER submission. SEER indicates Surveillance, Epidemiology, and End Results.

decline, with the rate of decline slowing for white females in 2011 (APC = 1.08%, not significant). The rate of decline in black females has become more pronounced since 2007 (APC = -3.4%, significant) and appears to be continuing. Male CRC trends should be watched to determine whether the rates continue to plateau or change direction. The changes in the trend for CRC are varied

and depend on the race/sex group; they mostly indicate a decline in the rate. It will be important to monitor these trends in light of CRC screening colonoscopy participation^{9,10} and CRC rates among the young.¹¹

Other trends to watch include the decline in lung and bronchus cancer. Trends for white males and white females are declining, although the rate of decline has slowed for

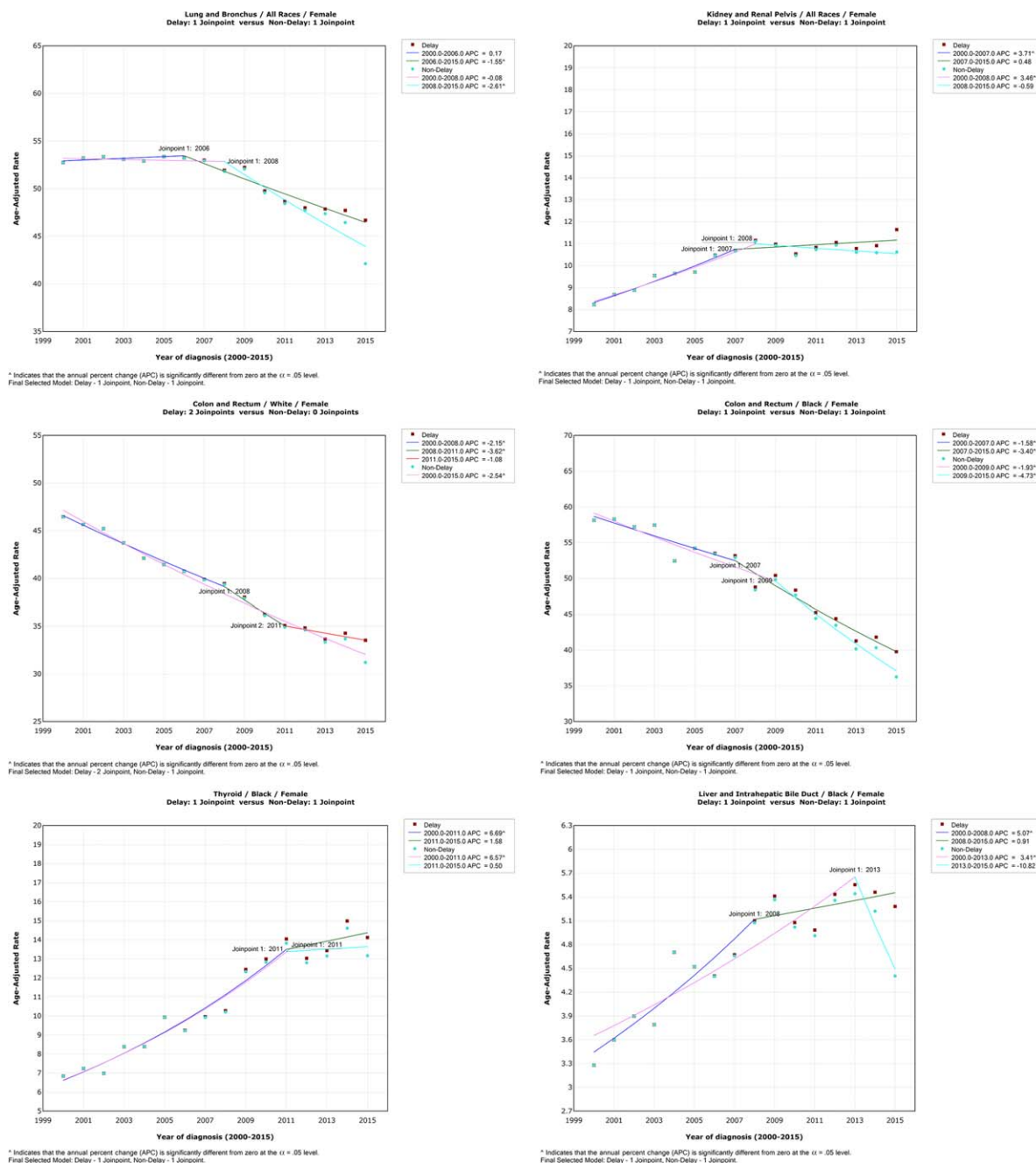


Figure 2. SEER 18 registries' observed and delay-adjusted incidence rates for selected sites for females for the diagnosis years 2000-2015 from the February 2017 SEER submission. SEER indicates Surveillance, Epidemiology, and End Results.

each in the last few years (APC for white males = -2.29% , significant, 2012-2015; APC for white females = -0.73% , significant, 2011-2015). Lung and bronchus cancer shows a continuous decline for black males (APC = -2.95% , significant, 2000-2015) and black females (APC = -1.47% , significant, 2005-2015). Cervical cancer among white females changed to a more pronounced decline over the interval for

the entire period under analysis, whereas the strong declining trend among black females is continuing; this may be a result of adhering to clinical guidelines for cervical cancer screening.¹²⁻¹⁴ The rising trend for female thyroid cancer among all races and white females diminished slightly. The rising trend for thyroid cancer among black females may have ended. Thyroid cancer incidence rates are known to

have risen markedly among US whites and blacks, and some portion of the rise may be real; however, overdiagnosis cannot be ruled out.¹⁵ Factors that may explain some portion of the increase in thyroid cancer may be changes in environmental factors, including noncurrent smoking and obesity.¹⁶ Obesity has been associated with more aggressive forms of papillary thyroid cancer.¹⁷⁻¹⁹ Liver/IBD cancer among black females appears to have had a noticeable change in trend from increasing for more than a decade to the introduction of a joinpoint, which has resulted in a flattening of the trend, whereas for black men, rates continue to rise. A recent report of National Center for Health Statistics and SEER data in the United States discusses increasing death rates of liver cancer among black Americans, and it indicates that the incidence among blacks may be reaching a plateau.²⁰ In our analysis, we did find evidence of a plateau among black females (APC = 0.91%, not significant, 2008-2015). More than half of liver cancers are due to potentially modifiable risk factors, including infections with hepatitis B and C viruses, excess body weight and obesity, smoking, and alcohol consumption.²¹⁻²³ The increasing incidence of liver cancer has been reported previously.²⁴ Among US Medicare recipients, metabolic disorders contribute the most to hepatocellular cancer, and they are followed by hepatitis C virus.²² It is important to note that population-attributable risk factors differ by race and ethnicity, with metabolic disorders having the greatest share among whites and Hispanics and hepatitis C virus having the largest share among blacks in the US Medicare population.²² Kidney and renal pelvis cancer among white males started to rise, and for white females, there was a slight (although not statistically significant) rise. Among black males, the final trend segment for kidney and renal pelvis cancer became flat, with a more pronounced increase between 2000 and 2009. The trend for black females was consistently flat. The differences in trends for whites and blacks for kidney and renal pelvis cancer may indicate a change point for this cancer in the United States. Previously, the kidney and renal pelvis cancer incidence was deemed to be stable among males and females; however, sex in combination with race was not evaluated.²⁵ Obesity is an important risk factor for kidney and renal pelvis cancer. A recent update of obesity trends from 2005 to 2014 indicates a significant positive linear trend for both non-Hispanic white and non-Hispanic black women.²⁶ Obesity trends are rising for US adults older than 20 years generally,²⁷ and this may partly explain some of the increases. Prostate cancer incidence trends continue to decline. The declining trend seems to be slowing somewhat for both blacks and whites.

In conclusion, this year's analysis of early estimates for cancer incidence in 2015 reveal some interesting changes in trends by sex and the new addition of race groups for whites and blacks. The addition of early estimates for white and black race provide an additional dimension to the ongoing pursuit of analyzing subpopulations to offer improved coverage of early estimates. This information can hopefully be used by the cancer control community to more effectively target these populations for attention to health awareness and prevention efforts.

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CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

AUTHOR CONTRIBUTIONS

Denise Riedel Lewis: Planning of the analysis, review and interpretation of the results, and preparation of the manuscript. **Huann-Sheng Chen:** Planning of the analysis, refinement of the method for the current analysis, completion of the analysis, review and interpretation of the results, and preparation of the manuscript. **Myles G. Cockburn:** Review and interpretation of the results. **Xiao-Cheng Wu:** Review and interpretation of the results. **Antoinette M. Stroup:** Review and interpretation of the results. **Douglas N. Midthune:** Refinement of the method for the current analysis. **Zhaohui Zou:** Completion of the analysis. **Martin F. Krapcho:** Completion of the analysis. **Daniel G. Miller:** Completion of the analysis. **Eric J. Feuer:** Planning of the analysis, refinement of the method for the current analysis, review and interpretation of the results, and preparation of the manuscript.

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